

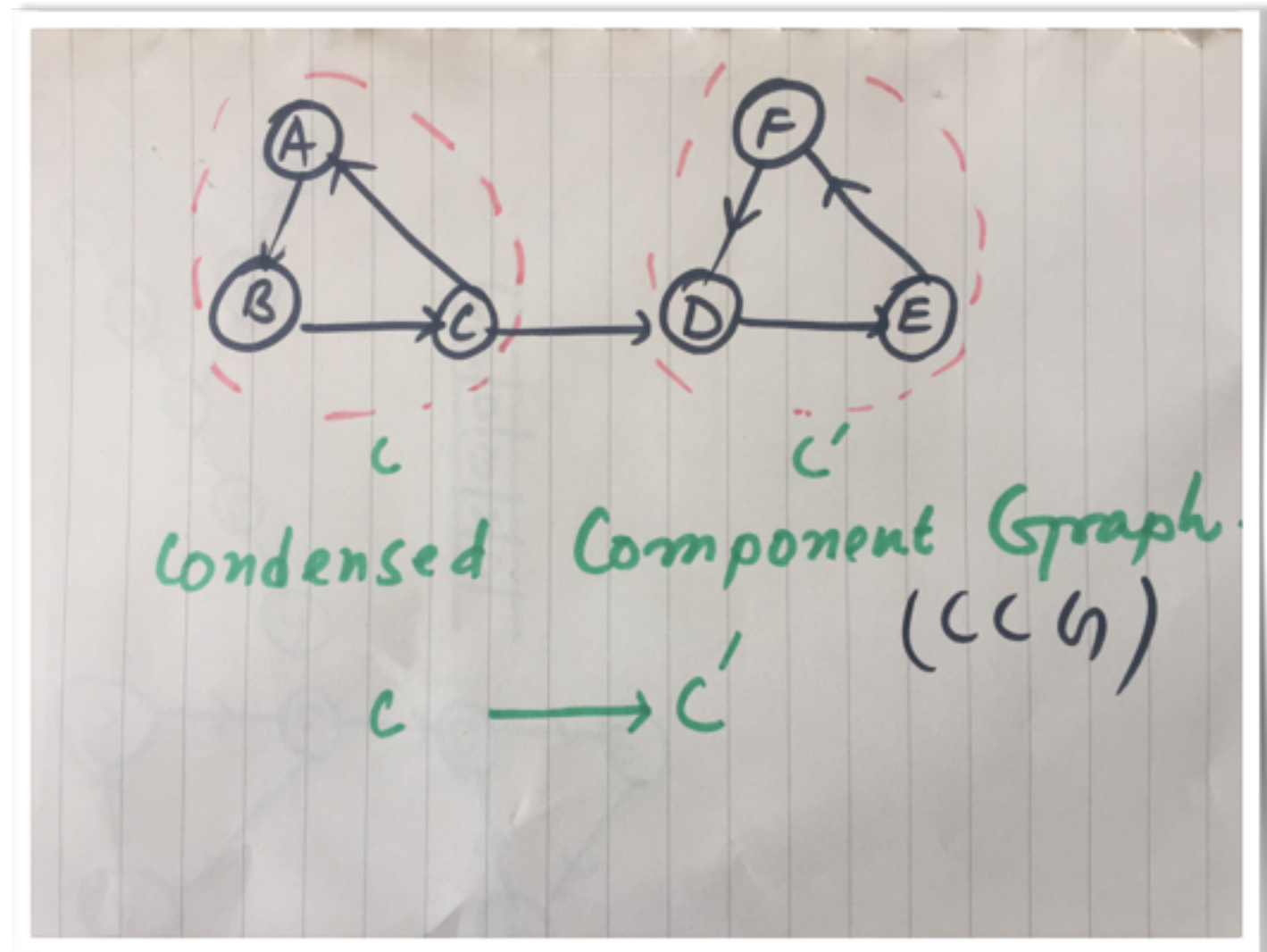
Kosaraju's algorithm for finding SCCs in a digraph

Debarka Sengupta

SCC

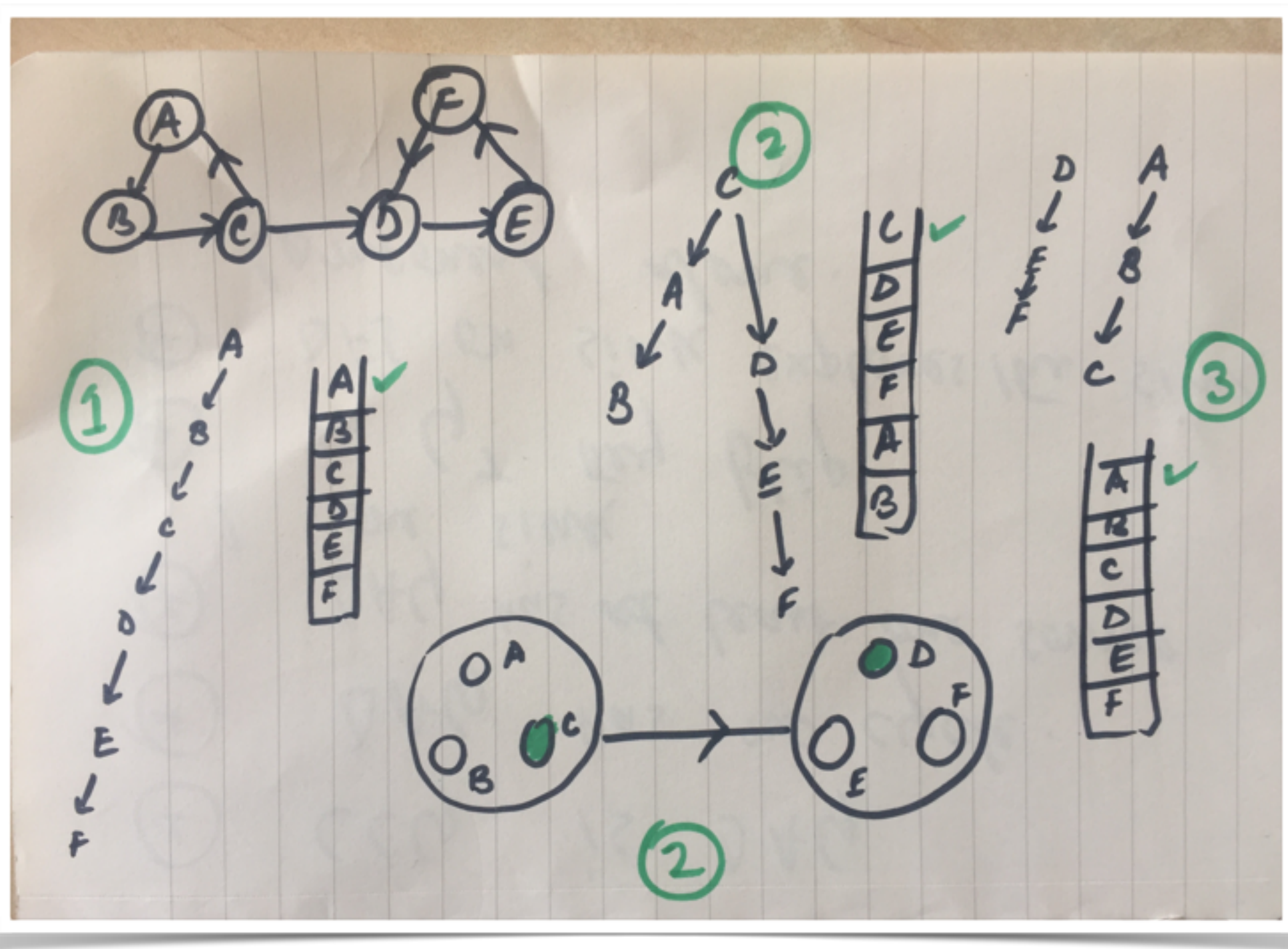
- Maximal subset of a graph with a directed path between any two vertices

Condensed component graph



- SCCs are treated as nodes
- CCG is a DAG - easy to show

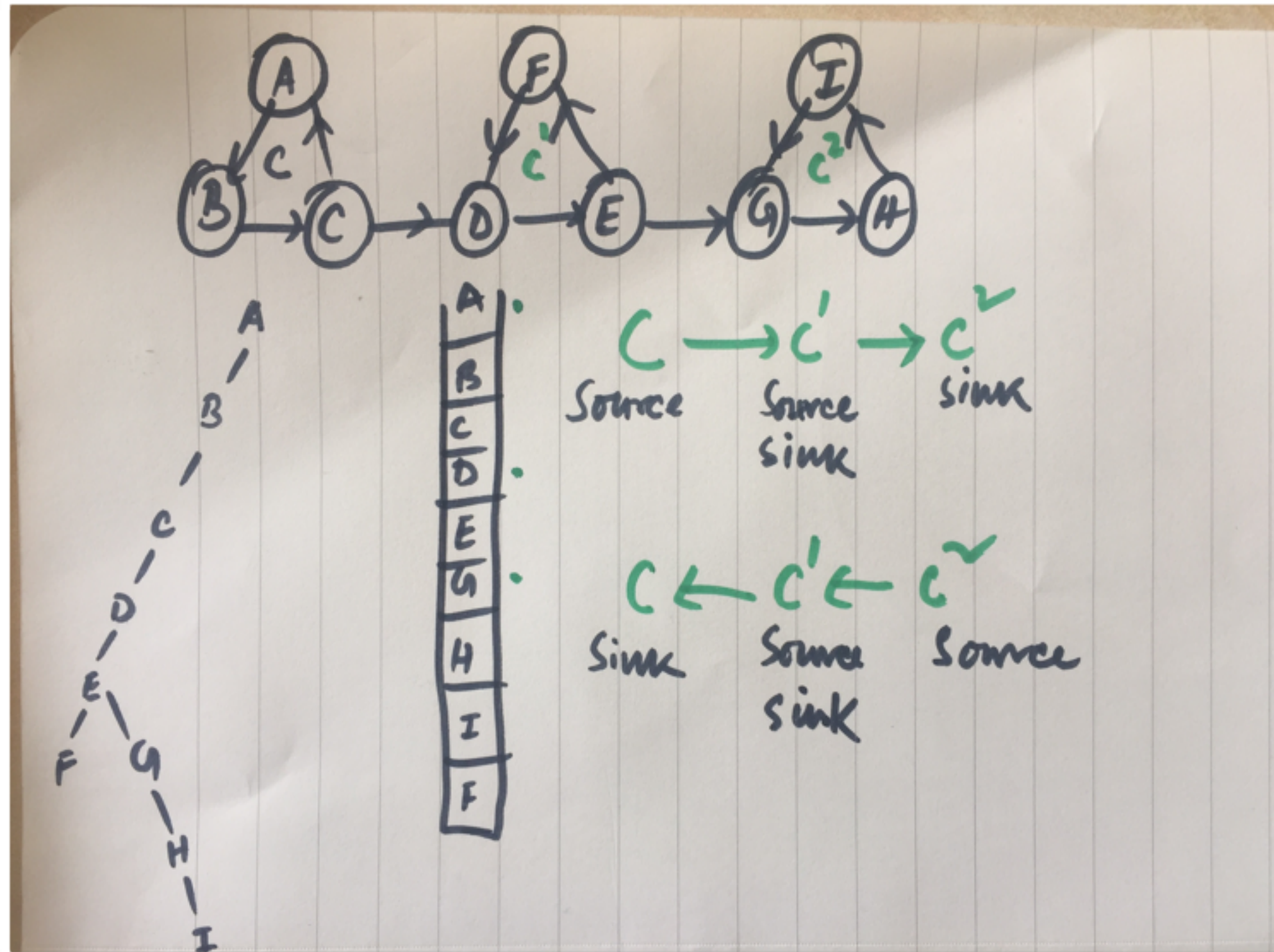
DFS on a digraph



Observation

- Now a property can be proven for any two nodes C^1 and C^2 of the Condensed Component Graph that share an edge, that is let $C^1 \rightarrow C^2$ be an edge. The property is that the finish time of DFS of some node in C will be always higher than the finish time of all nodes of C^2 .
- Check out <https://www.hackerearth.com/practice/algorithms/graphs/strongly-connected-components/tutorial/>

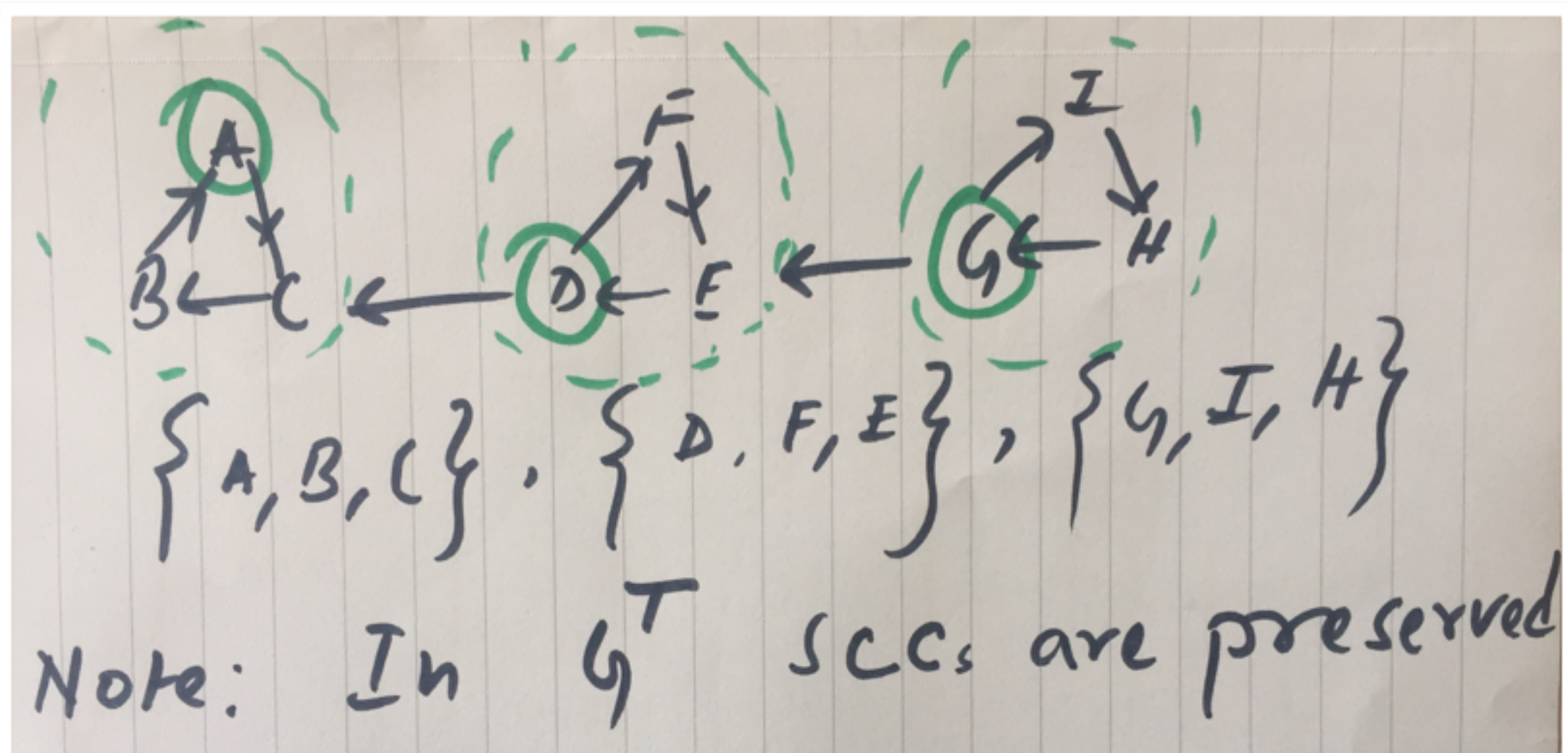
For a bigger graph



Ingest

- ① CCG is DAG
- ② DAG has no cycle.
- ③ DAG has at least one source
- ④ & one sink
- ⑤ In G^T they flip.
- ⑥ DFS on sink explores the sink component alone.

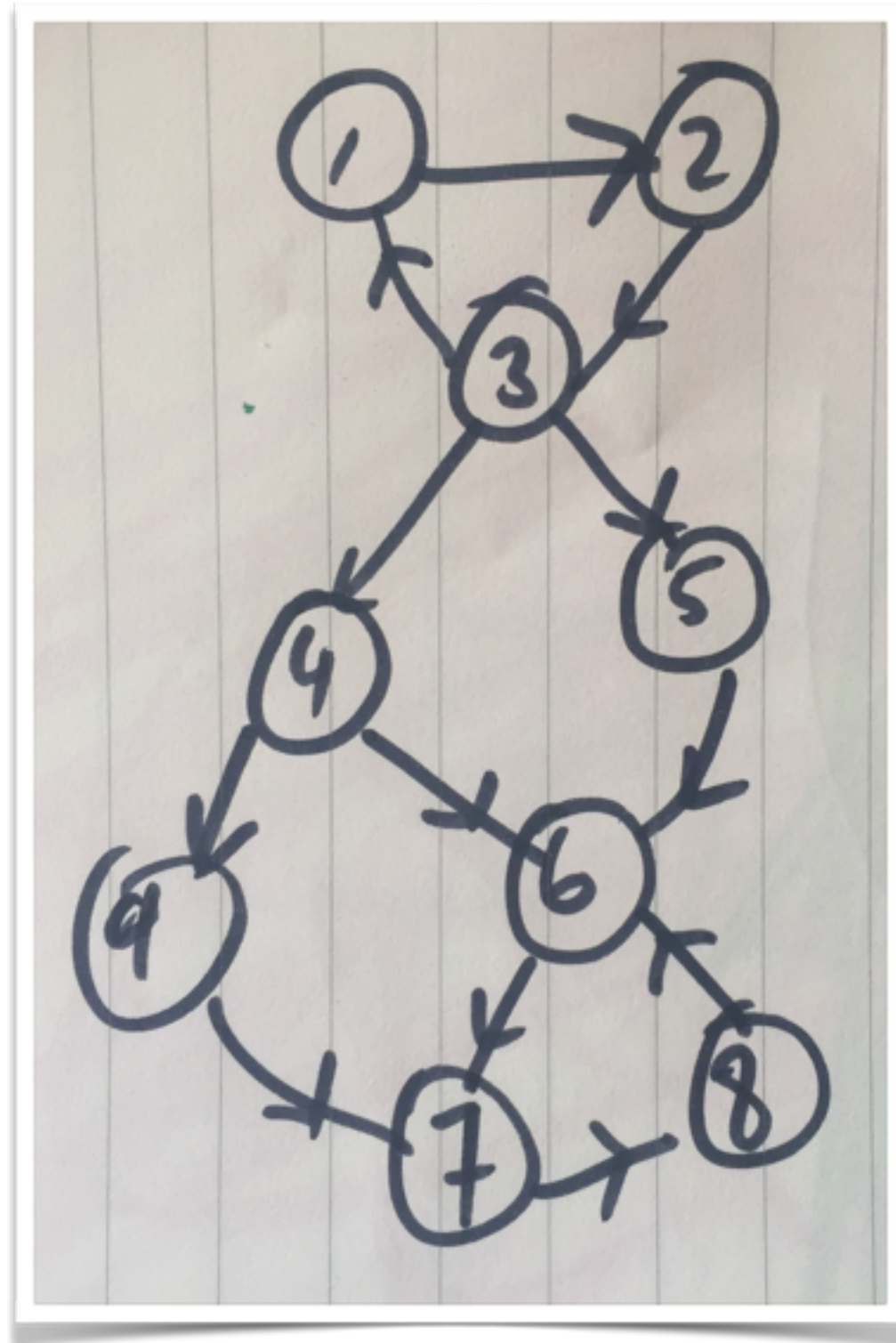
In transposed graph SCCs
are unaltered



So clever

- Start with top node in stack, which represents source component in G
- In G^T , that node represents a sink component
- DFS on from that node explores the entire sink component in G^T
- SCCs remain the same across G and G^T
- Removal of sink leaves CCG as a DAG only (can you prove)

Solve yourself



Solution

